PBL-III (AIM3170) • MENTOR PRESENTATION

AI-Powered UPI Scam & Fraud Detection

Behavioral Anomaly Detection & Multi-Hop Graph Layering Analytics

PROJECT DOMAIN
- FinTech Cybersecurity
- Unsupervised Anomaly Detection
- Directed Graph NetworkX

CORE ML ENGINE
- Isolation Forest Model
- StandardScaler Normalization
- Exact 30 Model Features

ACADEMIC INSTITUTION
- Manipal University Jaipur
- Dept. of AIML (2026)
- Application-Based Project

1. Problem Statement & Industry Motivation

Why traditional payment fraud rules fail in real-time UPI ecosystems

The Modern UPI Scam Challenge

- **Massive Velocity & Irreversibility:** Instant settlements give victims zero rollback window.
- **Social Engineering Scams:** Phishing QR codes, fake refund links, and remote access tools coerce authorized debits.
- **Account Takeover Bursts:** Unrecognized devices draining 100% balance during nocturnal hours (22:00 – 06:00).
- **Mule Layering Networks:** Stolen money is split across multiple intermediary accounts within minutes to prevent fund freezing.

Why Traditional Rule Engines Fail

- **High False Positive Rates:** Static rules (Amount > $50k) block genuine users while missing sophisticated micro-layering.
- **Siloed Per-Transaction View:** Evaluating transfers individually fails to detect multi-transfer bursts and rapid forwarding chains.
- **Lack of Explainability:** Black-box fraud scores lack interpretable behavioral context required for fast triage.
- **Our Core Approach:** Unsupervised Isolation Forest + 30 chronological behavioral features + NetworkX graph topology.

■ **Objective: Build an interactive batch investigation system that ingests monthly transaction datasets, computes 30 leak-free behavioral features, ranks suspicious transactions by anomaly score, and provides actionable triage evidence.**

2. Previous Semester (PBL-II) Recap & Transition

Building upon proven competencies in machine learning, user interfaces, and persistence

PBL-II Foundation (AI Health Assistant)

- **Supervised Classification:** Trained multi-label classification models on user symptom vectors.
- **Fuzzy Dietary Recommendation:** Token-sort fuzzy matching on structured food knowledge bases.
- **Multi-Tab Streamlit UI:** Responsive web dashboards with authentication and session states.
- **Dual Data Persistence:** MongoDB database with structured local JSON fallback.
- **Core Competency:** Model serialization, validation, and full-stack Python deployment.

PBL-III Evolution (UPI Scam Detection)

- **Supervised → Unsupervised:** Replaced static class labels with Isolation Forest to detect zero-day attacks without ground truth bias.
- **Single Input → Full Batch Analytics:** Ingesting full monthly transaction datasets rather than single form inputs.
- **Leak-Free Feature Engineering:** Engineered 30 temporal, rolling-window, and expanding statistics strictly prior to each event.
- **Relational Graph Modeling:** Added NetworkX graph layer to trace multi-hop layering chains (A → B → C → D).
- **Investigative Decision Support:** Tiered risk classifications and feature vector inspection.

3. End-to-End System Architecture

Reproducible, data-leak-free batch processing and inference pipeline

01. BATCH UPLOAD
- 14 Raw Columns
- Monthly Datasets
- Format Validation

02. DATA VALIDATION
- Nulls & Duplicate IDs
- Non-negative Values
- Datetime Parsing

03. FEATURE ENGINE
- Chronological Sorting
- 30 ML Features
- No Future Leakage

04. PREPROCESSING
- np.log1p (11 features)
- Scaler transform()
- Never refitted

05. ISOLATION FOREST
- Decision Score Output
- Predict: Normal/Anomaly
- Zero Retraining

06. STREAMLIT UI
- KPIs & Ranked Table
- 30-Feature Deep-Dive
- Scored CSV Export

■ **Architectural Rule: Pre-trained Isolation Forest and StandardScaler are completely immutable artifacts. No refitting or retraining occurs during inference.**

4. Feature Engineering: The 30 Model Features

Vectorized extraction strictly prior to current transaction to avoid future leakage

1. Direct & Temporal Features (10)
- amount, transaction_step, sender_old_balance, sender_new_balance
- receiver_old_balance, receiver_new_balance
- device_change_flag, new_device_flag
- transaction_hour (0–23), is_night (22:00–06:00)

2. Historical Interaction Counts (6)
- sender_txn_count_before (Prior sent txns)
- same_step_txn_count_before (Txns in same step)
- sender_receiver_txn_count_before (Pair history)
- sender_unique_recipients_before, receiver_txn_count_before, receiver_unique_senders_before

3. Expanding Statistics & Ratios (6)
- sender_balance_ratio (amount / [sender_old_balance + 1])
- time_since_previous_sender_txn (Hours, -1 if 1st txn)
- sender_amount_mean_before & sender_amount_std_before
- amount_deviation & amount_zscore (Normalized deviation)

4. Rolling Velocity & Chains (8)
- txns_last_1h, txns_last_6h, txns_last_24h (Rolling searchsorted)
- rapid_transaction_flag (Burst indicator: time <= 6min)
- hop_number (Position in transfer money flow chain)
- chain_length & chain_duration_steps, transaction_frequency

■ **Data Leakage Guarantee: Transactions are sorted chronologically. Every rolling window and cumulative count only evaluates events strictly prior to current event ($t_{prior} < t_{current}$).**

5. Interactive Batch Dashboard (Streamlit UI)

Operational investigation interface organized into 6 functional sections

Section 1: Upload & Validation
- Drag-and-drop CSV uploader
- Schema check on 14 raw columns
- 1-click Demo Dataset loader

Section 2: Overview KPIs
- Total Volume & Anomaly Rate (%)
- Min/Max Decision Scores
- Highest-Risk Transaction ID

Section 3: Suspicious Table
- Sorted by Anomaly Score ascending
- Filters: Risk Tier, Prediction, Type
- Limits: Top 10, 25, 50, 100, All

Section 4: Visual Analytics
- Normal vs. Anomaly breakdown
- Continuous score histogram
- Step trends & Transaction types

Section 5: Feature Inspector
- Single transaction deep-dive
- Full 30-feature vector viewer
- Raw vs. Transformed values

Section 6: Scored Exports
- Complete Scored Dataset CSV
- Anomalies-Only Filtered CSV
- Audit-ready data structures

■ **Risk Tiers: High Risk (Score < -0.05) | Medium Risk (-0.05 ≤ Score < 0.00) | Normal / Low Risk (Score ≥ 0.00)**

6. Automated Verification & Pipeline Reliability

Rigorous test suite verifying zero model drift, exact feature sequence, and export stability

15/15 Verification Suite (test_batch_app.py)

✓ **Test 1 & 2:** 500-transaction benchmark loaded & 14 columns validated.

✓ **Test 3, 4 & 5:** Exactly 30 features generated matching configuration order.

✓ **Test 6:** Exact 11 log features transformed using np.log1p().

✓ **Test 7 & 8:** StandardScaler.transform() executed; scaler was **NOT refitted**.

✓ **Test 9 & 10:** Isolation Forest loaded; model was **NOT retrained**.

✓ **Test 11 & 12:** 100% of transactions scored with decision score & discrete prediction.

✓ **Test 13, 14 & 15:** Original columns preserved; complete and filtered CSVs exported.

Live Detection Scenarios & Scoring

• **Normal Daytime Payment:**
Transfer $1,200 during daytime from primary device.
➔ **Score: +0.0237** | **Prediction: +1 (Normal / Low Risk)**

• **Nocturnal Account Takeover:**
Transfer $850k at 03:15 AM from new device draining 100% balance.
➔ **Score: -0.1428** | **Prediction: -1 (High Risk)**

• **Rapid Mule Layering Hop:**
Account receives transfer and forwards $79k downstream within 6 min.
➔ **Score: -0.0969** | **Prediction: -1 (High Risk)**

7. Post-Midterm Roadmap & Upcoming Milestones

Advancing from tabular behavioral anomaly detection to relational graph analysis

Milestone 1: NetworkX Directed Graph Engine [IN PROGRESS]
Construct directed money-flow graphs where accounts represent nodes and fund transfers form weighted edges with timestamp metadata.

Milestone 2: Multi-Hop Scam Chain Detection [UPCOMING]
Implement cycle detection and path search algorithms to identify rapid layering chains (A → B → C → D) and fund dispersal syndicates.

Milestone 3: Combined Multi-Signal Risk Engine [UPCOMING]
Synthesize behavioral Isolation Forest scores (60%) with graph topology centralities (40%) into a unified 0–100 composite risk score.

Milestone 4: Interactive Network Visualizer [UPCOMING]
Develop interactive node-link visualizers in the Streamlit UI to display money flow subgraphs for suspicious accounts.

Milestone 5: MongoDB Persistence & Final Benchmark [UPCOMING]
Integrate MongoDB cloud storage for audit trails, case notes, and comprehensive precision/recall evaluation against synthetic fraud benchmarks.

8. Project Summary & Mentor Discussion

Recap of progress achieved and open discussion points with mentor

Key Achievements Till Mid-Term

- ✓ **Working Batch Engine:** Built high-performance feature engineering calculating 30 ML features from raw monthly CSV batches without future data leakage.
- ✓ **Strict Model Integrity:** Integrated pre-trained Isolation Forest and StandardScaler as immutable artifacts with zero refitting or retraining.
- ✓ **Streamlit Dashboard:** Complete 6-section UI with KPIs, suspicious transaction filters, charts, deep-dive inspectors, and exports.
- ✓ **100% Automated Test Suite:** Passed all 15 operational test assertions validating pipeline correctness.

Discussion Points for Mentor Guidance

- ■ **Graph Topology Scaling:** Best practices for constructing dynamic transaction subgraphs in NetworkX when batch sizes scale past 100,000 nodes.
- ■ **Composite Risk Weighting:** Refining weighting formulas between tabular Isolation Forest anomaly scores (60%) and graph degree centralities (40%).
- ■ **Evaluation Benchmarks:** Feedback on synthetic fraud injection scenarios (layering, account takeover, fan-out dispersal) for benchmarking.
- ■ **Live Demonstration:** Ready to demonstrate the interactive Streamlit batch application during review.